

PRACTICAL 6

Named Pipes (FIFO) and POSIX Signal Handling

Question 1: Create a client-server application using Named Pipes (FIFOs). The client sends messages and the server processes and responds to them. Analyze FIFO behavior when multiple clients communicate simultaneously.

What we did

We created a simple client-server communication system using a named pipe called **myfifo**. The server waits for a message, reads the client's message, and sends a response back. The client sends **Hello Server** and receives the server response.

Commands used

Compile: gcc Server.c and gcc Client.c

Run: ./a.out

The server and client were run in separate terminals.

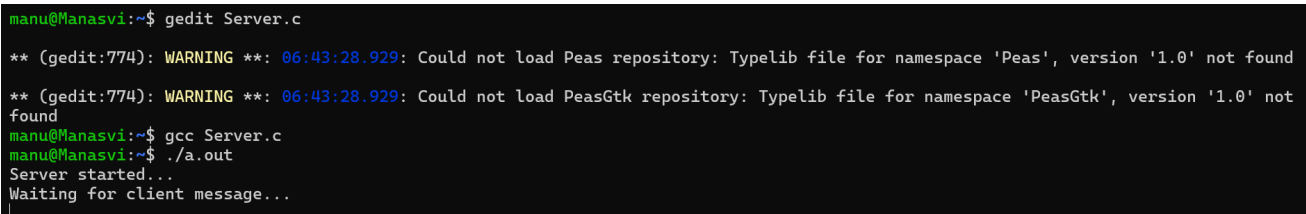
Output

Server: **Client message: Hello Server → Server response sent**

Client: **Message sent to server → Server response: Message processed by server**

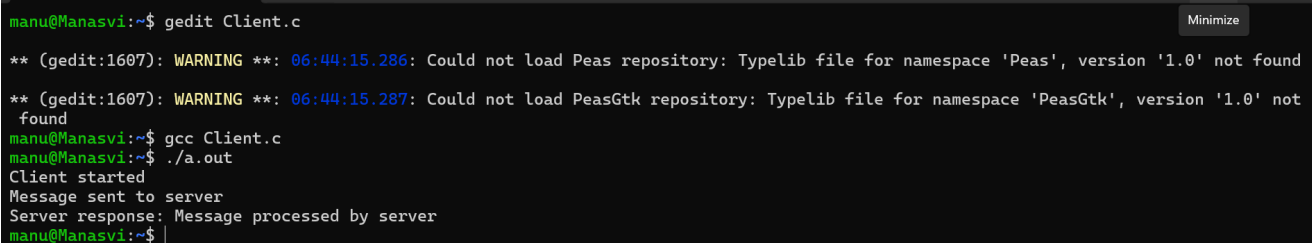
Observation

FIFO allows separate processes to exchange data. The server waits until the client sends a message, then processes it and sends a response. Multiple clients can use the FIFO, but message access is shared between them.



```
manu@Manasvi:~$ gedit Server.c
** (gedit:774): WARNING **: 06:43:28.929: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:774): WARNING **: 06:43:28.929: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
manu@Manasvi:~$ gcc Server.c
manu@Manasvi:~$ ./a.out
Server started...
Waiting for client message...
|
```

Screenshot 1 – FIFO execution



```
manu@Manasvi:~$ gedit Client.c
** (gedit:1607): WARNING **: 06:44:15.286: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:1607): WARNING **: 06:44:15.287: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
manu@Manasvi:~$ gcc Client.c
manu@Manasvi:~$ ./a.out
Client started
Message sent to server
Server response: Message processed by server
manu@Manasvi:~$ |
```

Screenshot 2 – FIFO execution

```

manu@Manasvi:~$ gedit Server.c

** (gedit:774): WARNING **: 06:43:28.929: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:774): WARNING **: 06:43:28.929: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
manu@Manasvi:~$ gcc Server.c
manu@Manasvi:~$ ./a.out
Server started...
Waiting for client message...
Client message: Hello Server
Server response sent
manu@Manasvi:~$ |

```

Screenshot 3 – FIFO execution

<pre> manu@Manasvi: ~ manu@Manasvi:~\$ gedit Server.c ** (gedit:774): WARNING **: 06:43:28.929: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found ** (gedit:774): WARNING **: 06:43:28.929: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found manu@Manasvi:~\$ gcc Server.c manu@Manasvi:~\$./a.out Server started... Waiting for client message... Client message: Hello Server Server response sent manu@Manasvi:~\$ </pre>	<pre> manu@Manasvi: ~ manu@Manasvi:~\$ gedit Client.c ** (gedit:1607): WARNING **: 06:44:15.286: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found ** (gedit:1607): WARNING **: 06:44:15.287: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found manu@Manasvi:~\$ gcc Client.c manu@Manasvi:~\$./a.out Client started Message sent to server Server response: Message processed by server manu@Manasvi:~\$ </pre>
---	--

Screenshot 4 – FIFO execution

Question 2: Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

What we did

We registered handlers for **SIGINT**, **SIGTERM**, and **SIGUSR1**. The program displays its process ID and waits for signals. Signals were sent from another terminal using the **kill** command. The handler prints the received signal. After all three signals are received, the program exits.

Commands used

Compile: gcc Signal.c

Run: ./a.out

Send signals: kill -SIGUSR1 PID kill -SIGTERM PID kill -SIGINT PID

Output

The program displayed the process ID, then showed **SIGUSR1 received**, **SIGTERM received**, and **SIGINT received**, followed by **All signals received. Exiting...**

Observation

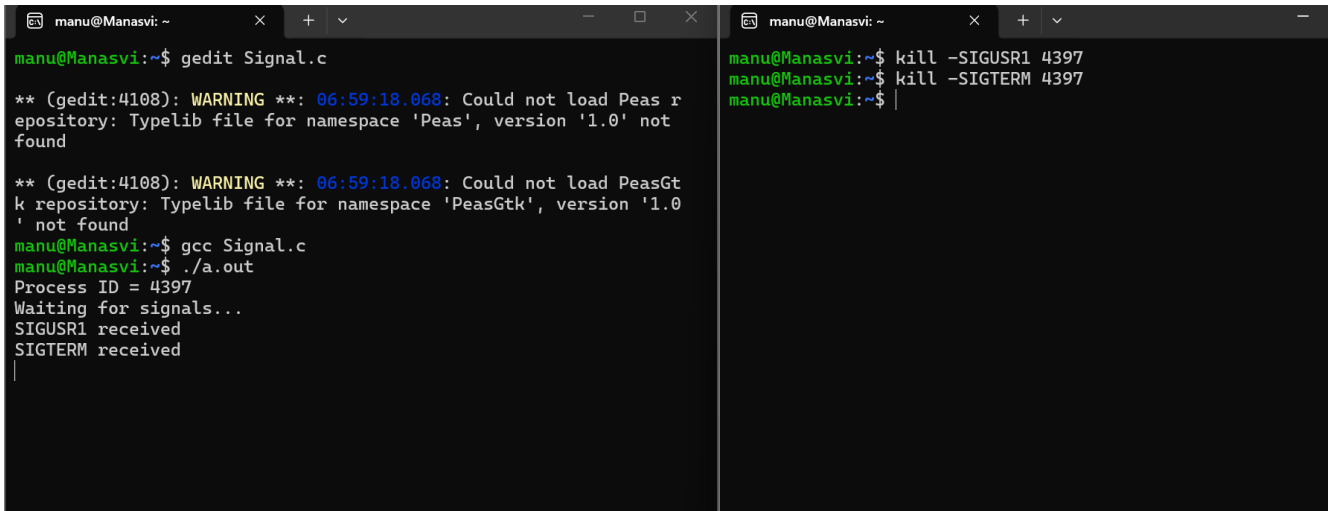
Signal handling lets a running process respond to asynchronous events. The signal handler identifies the signal and performs the required action without the program continuously checking for it.

```
manu@Manasvi:~$ gedit Signal.c
** (gedit:4108): WARNING **: 06:59:18.068: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:4108): WARNING **: 06:59:18.068: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
manu@Manasvi:~$ gcc Signal.c
manu@Manasvi:~$ ./a.out
Process ID = 4397
Waiting for signals...
```

Screenshot 5 – Signal handling execution

<pre>manu@Manasvi:~\$ gedit Signal.c ** (gedit:4108): WARNING **: 06:59:18.068: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found ** (gedit:4108): WARNING **: 06:59:18.068: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found manu@Manasvi:~\$ gcc Signal.c manu@Manasvi:~\$./a.out Process ID = 4397 Waiting for signals... SIGUSR1 received</pre>	<pre>manu@Manasvi:~\$ kill -SIGUSR1 4397 manu@Manasvi:~\$</pre>
---	---

Screenshot 6 – Signal handling execution



```
manu@Manasvi: ~$ gedit Signal.c

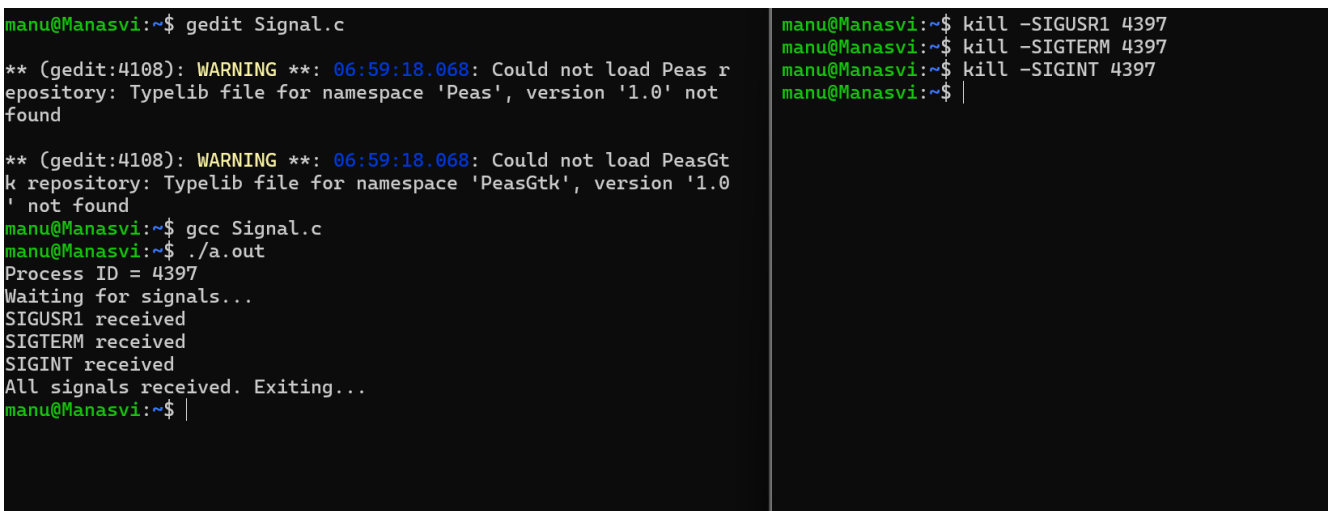
** (gedit:4108): WARNING **: 06:59:18.068: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:4108): WARNING **: 06:59:18.068: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

manu@Manasvi:~$ gcc Signal.c
manu@Manasvi:~$ ./a.out
Process ID = 4397
Waiting for signals...
SIGUSR1 received
SIGTERM received

manu@Manasvi:~$ kill -SIGUSR1 4397
manu@Manasvi:~$ kill -SIGTERM 4397
manu@Manasvi:~$
```

Screenshot 7 – Signal handling execution



```
manu@Manasvi:~$ gedit Signal.c

** (gedit:4108): WARNING **: 06:59:18.068: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:4108): WARNING **: 06:59:18.068: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

manu@Manasvi:~$ gcc Signal.c
manu@Manasvi:~$ ./a.out
Process ID = 4397
Waiting for signals...
SIGUSR1 received
SIGTERM received
SIGINT received
All signals received. Exiting...
manu@Manasvi:~$

manu@Manasvi:~$ kill -SIGUSR1 4397
manu@Manasvi:~$ kill -SIGTERM 4397
manu@Manasvi:~$ kill -SIGINT 4397
manu@Manasvi:~$
```

Screenshot 8 – Signal handling execution

Result: Successfully implemented FIFO client-server communication and POSIX signal handling and verified both programs through terminal execution.